International Journal of Engineering, Science and Information Technology
Volume 6, No. 1 (2026) pp. 110-115
eISSN: 2775-2674
Website: http://ijesty.org/index.php/ijesty
DOI: https://doi.org/10.52088/ijesty.v6i1.1825
Research Paper, Short Communication, Review, Technical Paper



AI Governance Across the Structured-to-Fluid Automation Spectrum: Deterministic and Agentic Architectures in Modern Contact Centres

Vikas Prasad*

Independent Researcher, United State

**Corresponding author Email: reachvikaspd@gmail.com*

The manuscript was received on 11 May 2025, revised on 29 August 2025, and accepted on 29 December 2025, date of publication 21 January 2026

Abstract

Determining the appropriate role of artificial intelligence (AI) within enterprise architecture presents a greater challenge than the technical deployment of AI itself. Modern contact centres operate across a wide spectrum of customer interactions, ranging from highly structured transactions that require strict compliance and auditability to complex conversations that demand contextual understanding and adaptive decision-making. This article proposes a governance-driven architectural framework based on a structured-to-fluid automation spectrum, which maps the operational characteristics of service interactions to the most appropriate automation and AI capabilities. Rather than adopting a technology-first approach, the framework emphasizes governance as the primary design principle, focusing on acceptable levels of operational variance, regulatory risk exposure, and the preservation of human accountability. The proposed framework integrates deterministic workflow execution, hybrid guardrail architectures, AI interpretation layers, asynchronous channel governance, agent augmentation capabilities, anomaly detection mechanisms, and comprehensive quality evaluation into a unified operational model. By positioning each interaction type along the automation spectrum, organizations can systematically determine where rule-based automation, human oversight, and AI-driven reasoning should be applied. This approach enables enterprises to balance efficiency, customer experience, compliance requirements, and operational resilience while reducing the risks associated with uncontrolled AI adoption. To support practical implementation, the framework incorporates a crawl–walk–run maturity progression that guides organizations through incremental stages of AI adoption. Enterprises can begin with tightly governed automation, expand toward AI-assisted decision-making, and ultimately evolve into agentic systems as governance capabilities mature. The framework provides a structured pathway for integrating AI into contact centre operations while maintaining the reliability, transparency, and accountability required in mission-critical service environments.

Keywords: *Artificial Intelligence Governance, Deterministic Systems, Hybrid Automation, Agentic Architecture, Contact Centre Strategy.*

1. Introduction

Few decisions carry more long-term consequences in enterprise architecture than choosing between adaptive and deterministic systems. Artificial intelligence has expanded what contact centres can do, enabling nuanced interpretation, real-time sentiment tracking, and multi-step orchestration that rigid scripting could never approximate [1]. Yet the operational realities of regulated service environments have not changed. Billing disputes, identity verification, and prescription processing demand traceability, consistency, and zero tolerance for variance [2]. Deploying a probabilistic model where a deterministic one is warranted does not represent progress; it introduces risk.

What is missing from most automation strategies is not capability; it is a principled basis for deciding which capability belongs where. Organizations often select technology based on what is available or what peers have adopted, rather than on a careful analysis of what each use case requires. This technology-first tendency produces predictable failure patterns: adaptive systems deployed in compliance-sensitive contexts, where their probabilistic nature cannot be audited, and rigid scripts applied to fluid customer scenarios, where they frustrate rather than serve.

The governance framework introduced here directly addresses that gap. By framing automation as a spectrum rather than a binary choice, it gives architects and operations leaders a shared vocabulary for evaluating any service interaction, synchronous or asynchronous, self-service or agent-assisted, and selecting an architecture that matches its operational profile. The sections that follow build this argument from the ground up: starting with the distinct roles of deterministic and artificial intelligence systems, establishing the spectrum construct, then working through its application in email governance, agent augmentation, quality evaluation, and phased adoption.

Copyright © Authors. This is an open access article distributed under the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

2. Literature Review

2.1. Deterministic Systems: Precision and Compliance

There is a category of service transaction where determinism is not a design preference; it is a requirement. When every valid input is enumerable, every decision rule is fixed, and every outcome must survive regulatory scrutiny, probabilistic systems introduce risk without adding value [3]. The audit trail that a deterministic workflow produces, which is reproducible, inspectable, and formally verifiable, is something that even a well-performing machine learning model cannot replicate with equivalent reliability.

This matters in settlement systems as much as it does in customer-facing workflows. Distributed financial infrastructure, for instance, demands correctness guarantees across multi-zone deployments, where autonomous replication architectures enforce zero-loss commit governance and deterministic convergence to prevent divergence from cascading into settlement-impacting inconsistencies [4]. The same principle applies to contact centre transactions: when a billing update or account change is executed, the outcome must be identical regardless of which agent, channel, or system processed it.

Password resets, prescription refills, structured booking modifications, and financial transfers—each of these scenarios has a finite input space and a bounded set of acceptable outcomes. Deploying artificial intelligence in these contexts does not make them more capable; it makes them less governable. A key discipline in governance-first design is resisting the pressure to apply adaptive systems simply because the technology is available. When the process logic is fixed and the inputs are known, deterministic execution remains the architecturally superior choice.

2.2. Artificial Intelligence Systems: Interpretation and Adaptability

Not every customer statement maps cleanly to a process. A message expressing frustration about a recurring charge, or a call from someone who cannot articulate exactly what they need, cannot be resolved by pattern-matching against a predefined workflow tree. These are the scenarios where artificial intelligence earns its place, not as a replacement for structured execution, but as the interpretation layer that makes structured execution possible when inputs are ambiguous [5].

Large language models and pre-trained transformers have dramatically expanded what interpretation looks like in practice. Intent classification, entity extraction, sentiment trajectory analysis, and next-best-action recommendation are now achievable at production scale. The challenge is not capability; it is knowing what to do with the output. Retrieval-augmented generation offers a partial answer: by grounding model responses in enterprise knowledge bases and verified customer records rather than relying solely on parametric weights, it limits the range of things a model can confidently assert. It reduces the risk of plausible but inaccurate outputs in high-stakes interactions [6].

The architecture that emerges from these constraints is a two-layer design. Artificial intelligence handles interpretation; it understands what the customer is asking, in what emotional state, and with what context. Deterministic components handle execution; they validate candidate actions against policy rules before any consequential action occurs. This separation keeps the adaptability of artificial intelligence from bleeding into the compliance-critical execution layer, while allowing each component to evolve independently.

3. Methods

3.1. Artificial Intelligence-Augmented Human Agents

Human agents remain essential at the fluid end of the spectrum, particularly for emotionally sensitive or high-stakes interactions where accountability must rest with a person. The question is not whether to involve agents, but how to provide them the information and support they need to perform well under cognitive load [11]. An agent receiving an escalated email thread with twenty prior exchanges, involving multiple issue types and a deteriorating sentiment trajectory, faces a meaningful challenge before typing a single word in response.

Artificial intelligence can absorb much of that preparation burden. Summarizing the interaction history, identifying the core unresolved requests, highlighting where sentiment shifted, and surfacing a response draft grounded in current policy each reduces the time an agent spends reconstructing context and increases the time available for actual resolution. The governance constraint is firm, however: every recommendation must be attributable, editable, and overridable without friction [12]. Agents are not reviewing outputs for accuracy; they are accountable for the responses they send, and the system must make that accountability real rather than nominal.

There is an underappreciated governance benefit to augmented agent workflows. When agents consistently modify or reject artificial intelligence recommendations in a particular scenario, this pattern signals that the model may be missing domain knowledge, the knowledge base may be outdated, or a policy may have changed without being reflected in the training data. Capturing override patterns and routing them into improvement pipelines turns the augmentation layer into a continuous feedback mechanism rather than a static capability.

3.2. Comprehensive Quality Evaluation and Anomaly Detection

Sampling-based quality evaluation has a structural problem that volume cannot solve it leaves many interactions unreviewed, creating blind spots that are invisible until they become incidents. A compliance failure pattern concentrated in a specific interaction type, or a sentiment deterioration trend affecting a particular customer segment, will not surface in a five-percent sample with any reliability [13]. By the time it does appear, the systemic issue behind it may have been operating for weeks.

Full-coverage evaluation eliminates that structural blind spot. When every self-service and human-assisted interaction is assessed against compliance, resolution quality, and customer experience criteria, the governance model changes qualitatively, not just in scale but in what it can detect. Anomaly detection extends the concept by operating in real time rather than retrospectively. Escalation rate spikes, misclassification increases following a model update, and sentiment deterioration correlated with a specific product line—these signals become actionable before they cascade into customer-impacting failures.

Telemetry-driven predictive frameworks take this capability even further. Rather than waiting for a quality or anomaly signal to emerge from interaction data, they convert raw infrastructure metrics, latency distributions, buffer cache pressure, and replication synchronization patterns into early-warning indicators that surface failure precursors before customers are affected [14]. The combination of full-coverage quality evaluation, real-time anomaly detection, and infrastructure telemetry creates a continuous governance loop with genuinely predictive rather than merely reactive properties.

Table 1. Anomaly detection signal categories and governance response obligations [13][14]

Signal category	Detection source	Governance response
Escalation rate deviation	Interaction volume telemetry	Model validation and confidence threshold review
Sentiment deterioration trend	Real-time conversation scoring	Routing policy adjustment and agent notification
Intent misclassification increases	Post-interaction classification audit	Knowledge base update and model retraining trigger
Infrastructure latency anomaly	Telemetry buffer and replication metrics	Capacity assessment and failover protocol activation
Policy override concentration	Agent override pattern analysis	Knowledge base synchronization and retraining pipeline trigger

3.3. Original Methodological Contribution and Field Advancement

The central contribution of this framework is its reframing of automation selection as a governance decision rather than a technology decision. To the best of the author's knowledge, no prior framework has unified deterministic, hybrid, and agentic systems under a single governance-based decision model applicable across synchronous and asynchronous service interactions. Prior treatments of enterprise automation have generally addressed deterministic and adaptive systems as separate domains, producing guidance optimized for individual deployment scenarios without a shared decision model applicable across the full range of service interaction types [17]. The spectrum construct provides a shared model that serves as a single governance lens for any use case, channel, or point along an organization's automation maturity curve.

The unification of deterministic execution, hybrid guardrails, and agentic orchestration within a single decision framework carries practical significance beyond conceptual tidiness. It means that the same governance principles governing a password reset workflow also govern a complex complaint resolution pathway; the criteria differ in outcome, but the design discipline is identical [18]. It also means that governance decisions made early in an organization's automation journey do not need to be discarded as the architecture matures; they scale along the spectrum as organizational confidence and infrastructure capability develop.

Elevating asynchronous channels, specifically electronic mail, from an implementation edge case to a demonstrative proof of the spectrum model is a further contribution. Email has historically been treated as a channel-specific problem, requiring channel-specific solutions. The framework shows that the same operational characteristics that govern synchronous automation decisions also apply to asynchronous interactions, simplifying the governance architecture considerably.

This work proposes a unified governance model that enables consistent decision-making across heterogeneous interaction types, a capability not addressed in the existing automation literature.

3.4. Broader Industry Implications

The implications of spectrum-based governance extend well beyond contact centre operations. Any enterprise domain characterized by heterogeneous transaction types, varying degrees of process rigidity, regulatory exposure, and input ambiguity faces the same fundamental architectural challenge [19]. Interaction populations that span the structured-to-fluid continuum are presented by financial services back-office operations, healthcare prior authorization workflows, insurance claims intake, and public sector service delivery, and each of these domains would benefit from the same governance-first classification discipline.

Regulatory pressure is moving in a direction that makes this framework increasingly relevant rather than optionally useful. The EU Artificial Intelligence Act and similar regulatory developments in other jurisdictions are establishing audit and explainability requirements that many currently deployed adaptive systems cannot satisfy [20]. Enterprises that have embedded governance-first design, where compliance-critical actions remain within deterministic execution layers and adaptive systems operate within explicitly defined bounds, will find regulatory compliance substantially more achievable than those that retrofit governance onto existing deployments. The framework described here positions organizations to meet those requirements by design rather than by remediation.

4. Results and Discussion

4.1. Structured-to-Fluid Automation Spectrum

The structured-to-fluid spectrum reframes automation selection as an evaluation of operational characteristics rather than a technology shortlist. At the structured end, interactions are rule-bound and fully determinable in advance. At the fluid end, they require adaptive reasoning, contextual judgment, and multi-step orchestration that no static script could anticipate. This spectrum also serves as a governance lens: interactions toward the fluid end entail greater ambiguity, higher regulatory exposure, and lower tolerance for outcome variance, factors that must inform architectural decisions alongside operational fit. Most real contact centre interactions fall somewhere between these poles, and the architecture must reflect where [7].

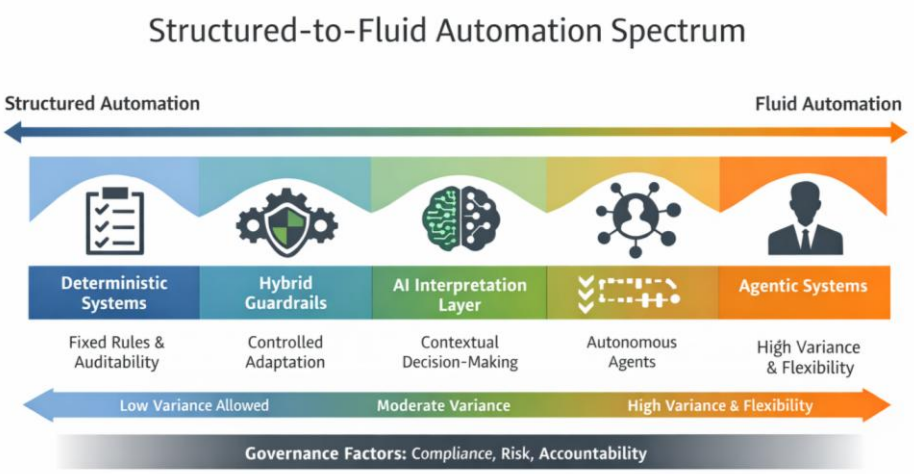


Fig 1. Automation Spectrum from Structured Systems to Flexible Agent-Based Automation

The governance value of the spectrum is that it forces the design conversation to start with the right questions. How much ambiguity is present in the input? How significant is the regulatory exposure if the outcome is wrong? What is the tolerance for outcome variance across the customer population? These questions yield more durable architectural decisions than asking which vendor's platform the organization has already licensed [8].

Table 1. Integrated Automation Spectrum Decision Matrix

Highly Structured	Bill payment, password reset, prescription refill, and booking confirmation	Deterministic Workflow	Validation, auditability, and compliance
Structured with Exceptions	Returns; refunds; account updates; warranty intake	Hybrid (AI Interpretation + Deterministic Execution)	Guardrail enforcement and exception handling
Mixed Emotional	Escalation requests, dissatisfaction, and policy negotiation	Hybrid with sentiment-aware triage	Consistency monitoring and risk review
Fluid Diagnostic	Outage support; troubleshooting; multi-step diagnostics	AI-Orchestrated System	Model oversight, safe action bounds
Highly Fluid Personalized	Complaint resolution, retention, and personalized recommendations	Agentic AI Experience (bounded)	Bias, drift, and safety governance

Table 1 translates the spectrum into a working decision tool. Each row represents a distinct operational archetype, not just a technology selection but a governance commitment. Hybrid designs in the middle range of the spectrum are architecturally distinct from both poles: artificial intelligence interprets and proposes, while deterministic components validate and execute. At the agentic end, the governance challenge intensifies considerably. Bounded agentic systems require explicit action-space constraints, defined confidence thresholds below which escalation is triggered, and runtime enforcement of safety bounds, none of which can be left to the model to self-govern.

4.2. Asynchronous Interaction Governance: Electronic Mail as a Spectrum Case Study

Electronic mail demonstrates, within a single channel, exactly why a spectrum-based governance model is necessary. One inbound message might be a structured refund confirmation request, with every required field present, intent unambiguous, and resolution deterministic. The next might be a nuanced complaint about a disputed charge, written with emotion and incomplete information, requiring interpretation, policy judgment, and contextual reasoning before any response is possible [9].

Treating these two messages with the same architecture is a governance failure in either direction. Applying a deterministic workflow to a complex complaint produces a rigid, unhelpful response. Routing a straightforward status request through a generative model introduces unnecessary latency and unpredictability where none is warranted. The spectrum framework resolves the issue of message classification by classifying messages based on operational characteristics rather than the channel before routing them to the appropriate processing pathway.

In practical terms, this means artificial intelligence performs upstream triage: classifying intent, detecting sentiment, and extracting the entities a downstream workflow will need. If required fields are missing, an automated request for missing information is sent before any process step is attempted. If the message requires policy discretion, it is routed through a fluid pathway with human or agentic oversight. Deterministic guardrails enforce compliant execution wherever the pathway leads to a structured outcome [10]. The same governance logic that applies in synchronous self-service channels transfers directly to email, with no channel-specific policy redesign required.

Table 2. Governance oversight obligations by interaction channel and automation tier [2], [8]

Interaction channel	Automation tier	Primary oversight obligation
Synchronous voice	Highly structured	Audit trail enforcement and deterministic rule validation
Synchronous voice	Fluid agentic	Real-time sentiment monitoring and escalation routing
Asynchronous electronic mail	Structured	Intent triage, field verification, and compliant routing
Asynchronous electronic mail	Fluid or mixed	Policy judgment review and human agent oversight
Omnichannel self-service	Mixed spectrum	Guardrail enforcement and systematic override capture

4.3. Crawl–Walk–Run Adoption Framework

Governance maturity cannot be installed all at once. Organizations that attempt to deploy bounded agentic orchestration before establishing foundational governance infrastructure, audit logging, validation rules, escalation pathways, and oversight processes discover that the complexity of agentic systems amplifies any governance gaps beneath them [15]. A phased progression is not merely cautious; it is architecturally sound.

The crawl phase establishes that foundation. Deterministic automation for highly structured scenarios, the right end of the spectrum, builds the operational discipline and technical infrastructure that subsequent phases depend on. Organizations learn what governance looks like in practice: how audit trails are structured, where escalation boundaries sit, and how exceptions are handled. The walk phase introduces artificial intelligence interpretation alongside deterministic guardrails for mixed scenarios, enabling the interpretation layer to be validated against the existing governance baseline. Override patterns inform model improvement; edge cases refine guardrail logic [16].

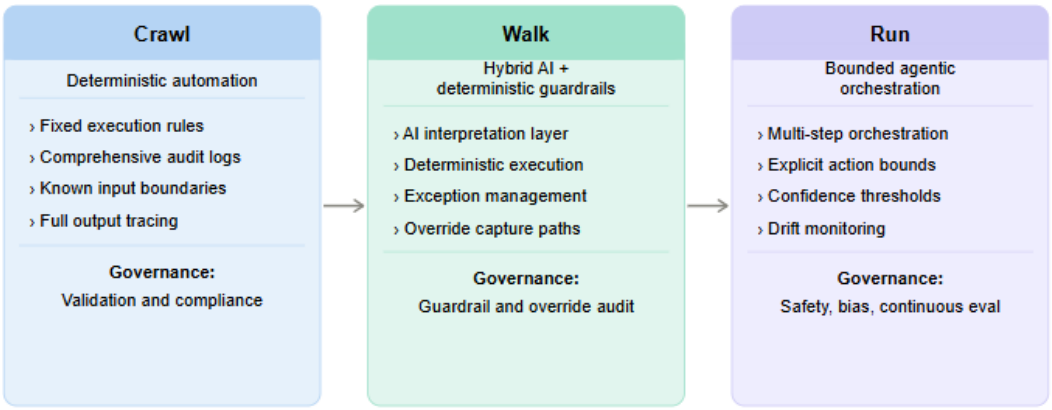


Fig 2. Crawl–Walk–Run Maturity Model

The run phase, bounded agentic orchestration for highly fluid scenarios, should only be entered when walk-phase baselines are stable. At this stage, governance design is most demanding: action spaces must be explicitly bounded, confidence thresholds must be defined, and safety constraints must be enforced through runtime controls rather than model self-regulation.

5. Conclusion

The governance-first framework presented across the preceding sections establishes a principled basis for enterprise automation design in contact centre environments, one that does not begin with a technology shortlist but with a disciplined evaluation of what each service interaction demands. The structured-to-fluid automation spectrum operationalizes this principle by providing architects and operations leaders with a shared decision model applicable across synchronous and asynchronous channels, self-service and human-assisted workflows, and at every point along an organization's automation maturity curve.

Three architectural conclusions emerge consistently from this framework. In compliance-sensitive, high-consequence contexts where outcome variance cannot be tolerated, and auditability is a regulatory obligation, deterministic execution remains irreplaceable. Artificial intelligence earns its architectural role as an interpretation and assistance layer, not as a wholesale replacement for structured execution but as the component that makes structured execution possible when inputs are ambiguous or emotionally complex. And agentic orchestration, at the fluid end of the spectrum, is only viable when it is bounded by explicit action constraints, confidence thresholds, and runtime safety enforcement that do not depend on model self-governance.

The governance infrastructure that supports these architectural choices, comprehensive quality evaluation at full interaction coverage, real-time anomaly detection, telemetry-driven predictive signals, and the phased crawl–walk–run adoption model transforms governance from a reactive audit function into a continuous operational discipline. Moving from sampling-based evaluation to full-coverage assessment represents a structural change in organizational accountability, not merely a scaling of existing practice.

As regulatory frameworks for enterprise artificial intelligence continue to mature globally, the architectural discipline described here, separating deterministic execution from adaptive interpretation, enforcing explicit bounds on agentic action, and maintaining observable governance loops across all interaction types, will increasingly define the baseline standard for responsible deployment. Enterprises that establish this discipline proactively, through governance-first design rather than post-deployment remediation, will be positioned to scale artificial intelligence capabilities without sacrificing the reliability, compliance confidence, and customer trust that mission-critical service environments require. This framework has implications for enterprise-scale AI governance and provides a foundation for future research and implementation in regulated environments.

References

[1] M.-H. Huang and R. T. Rust, "A strategic framework for artificial intelligence in marketing," *J. Acad. Mark. Sci.*, vol. 49, no. 1, pp. 30–50, 2021.

[2] D. Kalibatiene and O. Vasilecas, "A survey on modeling dynamic business processes," *PeerJ Comput—Sci.*, vol. 7, p. e609, 2021.

[3] M. Yang and S. Zhang, "Analysis of sports psychological obstacles based on mobile intelligent information system in the era of wireless communication," *Wireless. networks*, vol. 29, no. 8, pp. 3599–3615, 2023.

[4] F. Z. Gargab, "Technical-economic optimization and manufacture of a Moroccan solar water heater." Université de Pau et des Pays de l'Adour; Université Sidi Mohamed ben~…, 2021.

[5] B. Min *et al.*, "Recent advances in natural language processing via large pre-trained language models: A survey," *ACM Comput.*

Surv., vol. 56, no. 2, pp. 1–40, 2023.
[6] L. M. Amugongo, P. Mascheroni, S. Brooks, S. Doering, and J. Seidel, "Retrieval augmented generation for large language models in healthcare: A systematic review," *PLOS Digit. Heal.*, vol. 4, no. 6, p. e0000877, 2025.
[7] A. d'Avila Garcez and L. C. Lamb, "Neurosymbolic AI: The 3rd wave," *Artif. Intell. Rev.*, vol. 56, no. 11, pp. 12387–12406, 2023.
[8] V. K. Bollu, "Responsible AI governance in Enterprise Systems: A Risk and Compliance Framework," *Acta Sci*, vol. 27, p. 1, 2026.
[9] S. Pooja, G. Gokul, K. Linkesh Mani, A. S. Raj Kumar, and R. Amutha Bharathi, "Context-Aware AI Chatbot Using Transformer-Based Models for Intelligent User Interactions".
[10] S. Poria *et al.*, "Recognizing emotion cause in conversations," *Cognit. Comput.*, vol. 13, no. 5, pp. 1317–1332, 2021.
[11] Q. V. Liao and J. W. Vaughan, "AI transparency in the age of LLMS: A human-centered research roadmap," *Harvard Data Sci. Rev.*, no. Special Issue 5, 2024.
[12] A. Paleyes, R.-G. Urma, and N. D. Lawrence, "Challenges in deploying machine learning: a survey of case studies," *ACM Comput. Surv.*, vol. 55, no. 6, pp. 1–29, 2022.
[13] A. Rapp, A. Boldi, L. Curti, A. Perrucci, and R. Simeoni, "Collaborating with a text-based chatbot: an exploration of real-world collaboration strategies enacted during human-chatbot interactions," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–17.
[14] R. Gollapudi, "Telemetry-Driven Predictive Failure Models for High-Scale Financial Databases," *J. Comput. Anal. Appl.*, vol. 34, no. 12, 2025.
[15] D. ABHAYKUMAR, "Cognitive Robotic Process Automation (RPA) for Processing Unstructured Data," *Int. J.*, vol. 7, no. 2, pp. 639–643, 2022.
[16] E. Papagiannidis, P. Mikalef, and K. Conboy, "Responsible artificial intelligence governance: A review and research framework," *J. Strateg. Inf. Syst.*, vol. 34, no. 2, p. 101885, 2025.
[17] C. Trocin, P. Mikalef, Z. Papamitsiou, and K. Conboy, "Responsible AI for digital health: a synthesis and a research agenda," *Inf. Syst. Front.*, vol. 25, no. 6, pp. 2139–2157, 2023.
[18] J. Ledendal and others, "AI Data Governance: Overlaps Between the AI Act and the GDPR," *Law, Innov. Technol.*, 2026.
[19] W. H. Weiger, "Engaging business customers through online experiences in different cultures," *J. Int. Mark*, vol. 31, no. 3, pp. 59–79, 2023.
[20] M. Veale and F. Z. Borgesius, "Demystifying the draft EU artificial intelligence act," *arXiv Prepr. arXiv2107.03721*, 2021.